

Automated Plasma Exchange (Therapeutic Plasmapheresis)

What is automated plasma exchange?

Automated plasma exchange is a procedure to remove the liquid in blood (called plasma), which may contain harmful substances, and replace it with healthy fluid.

This process, also known as therapeutic plasmapheresis, involves withdrawing blood from a sick child. The blood is sent through a centrifuge in a blood cell separator apheresis machine, which separates plasma from the remaining blood cells. The plasma is discarded, and the remaining blood cells are mixed with replacement fluids and returned to your child.

If your child has an autoimmune disorder or received an organ transplant, your child's doctor may recommend plasmapheresis as part of his long-term care management.

Plasmapheresis is not a cure for your child's condition. However, it can help your child avoid complications from his disorder and improve his quality of life.

Who needs automated plasma exchange?

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work closely with specialists in Neurology, Nephrology, Hematology and Cardiology to develop plasmapheresis schedules that are customized to each patient's individual needs.

How is automated plasma exchange performed?

If your child is referred to the Apheresis Program for automated plasma exchange, he will have an initial evaluation by a physician-nurse team in the Apheresis Unit. The apheresis team will already know about your child and his condition from your child's treating physician.

On the day of your child's procedure, our team will greet you and your child when you arrive. You will be directed to a patient area that includes:

- A reclining chair or bed for your child
- A high-tech apheresis machine that will be customized for your child's size and procedure
- Chairs for you to sit with your child
- A television to distract and entertain your child

Your child's procedure will be overseen by board-certified physicians, and experienced apheresis nurses and Child Life staff. At least one team member will be with your child at all times to provide comfort, answer questions, and ensure the procedure goes smoothly.

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The blood that's withdrawn from your child will be circulated through a centrifuge in a blood cell separator machine. The centrifuge separates the plasma from the other parts of the blood (red blood cells, white blood cells, and platelets). The separated plasma is collected and discarded.

The remaining blood cells are mixed with replacement fluids (such as saline or albumin) or blood components (such as fresh frozen plasma), and returned to your child's vein. Blood and fluids are slowly removed and returned to your child at the same time, so your child is never missing more than a few ounces at a time.

Your child will also receive a small amount of citrate, an anticoagulant that will prevent your child's blood from clotting during the procedure.

An automated plasma exchange may take two to four hours to complete, depending on how much plasma is removed. The procedure may be repeated at regular intervals under the direction of your child's physician and the apheresis team. Depending on your child's condition, automated plasma exchange may be performed repeatedly for many years to help your child live a more normal life.

Risks

Automated plasma exchange is a safe procedure, but side effects can occur.

Your child may experience:

- Symptoms similar to regular blood donation, such as nausea, vomiting, dizziness or fainting
- Problems related to the anticoagulant, including chills, sour taste in the

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Contact the Apheresis Program

[Learn more about the Apheresis Program >](#)



Appointments and referrals

[215-590-1749](tel:215-590-1749)

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